

Sets

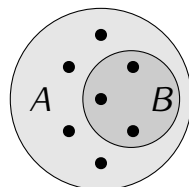
In general sets will be denoted by capital letters: $A, B, C, \dots$

Frequently used sets:

We will write $x \in A$ to denote that x is an element of the set A and $y \notin A$ to indicate that y is not an element of A .

Subsets

A set B is a *subset* of a set A if every element of B is in A . In such case we write $B \subseteq A$.



A set B is a *proper subset* of A if $B \subseteq A$ and $B \neq A$.

Example. Some subsets of $\mathbb{Z}$:

Operations on sets

- The *union* of sets A and B is the set $A \cup B$ that consists of all elements that belong to either A or B (or both):

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

- The *intersection* of sets A and B is the set $A \cap B$ that consists of all elements that belong to both A and B :

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

- The *difference* of sets A and B is the set $A \setminus B$ that consists of all elements that belong to A but not to B :

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$$

Example.

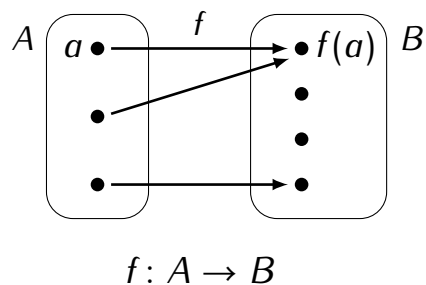
Note. We say that sets A and B are *disjoint sets* if $A \cap B = \emptyset$.

- The *Cartesian product* of sets A, B is the set consisting of all ordered pairs of elements of A and B :

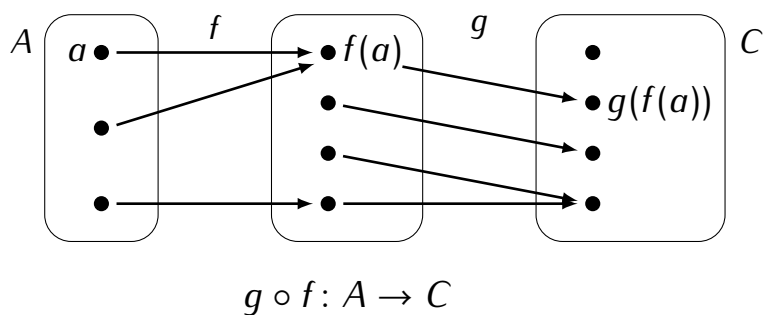
$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

Example.

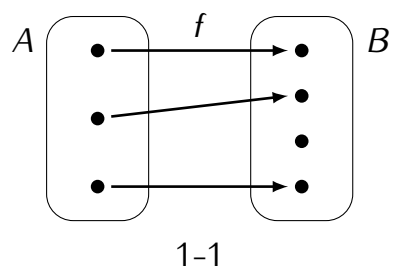
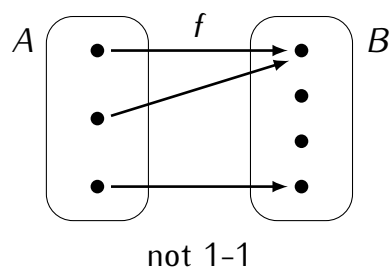
Functions



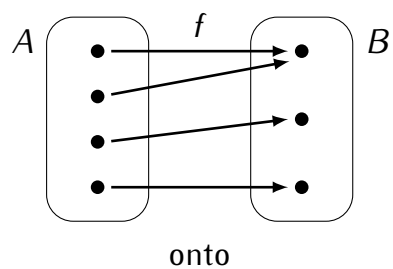
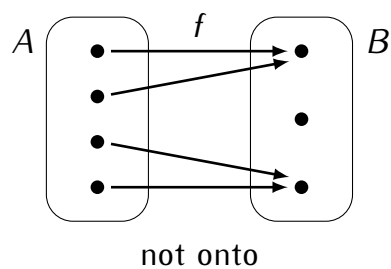
Composition of functions:



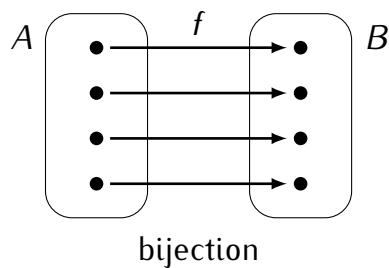
- A function $f: A \rightarrow B$ is *1-1* if $f(a) = f(a')$ only if $a = a'$.



- A function $f: A \rightarrow B$ is *onto* if for every $b \in B$ there is $a \in A$ such that $f(a) = b$.



- A function $f: A \rightarrow B$ is a *bijection* if f is both 1-1 and onto.



Note. If $f: A \rightarrow B$ is a bijection then the inverse function $f^{-1}: B \rightarrow A$ exists and it is also a bijection. Then for every $a \in A$ and $b \in B$ we have:

$$f^{-1}(f(a)) = a \quad \text{and} \quad f(f^{-1}(b)) = b$$

Cardinality

We will denote by $|A|$ the cardinality of the set A . For a finite set, this is the number of elements of A .

Note. If A, B are sets, then $|A| = |B|$ if and only if there exists a bijection $f: A \rightarrow B$.